

Food Science Labs





About the Course

Note that labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.

This topic is included in the following course(s): Science: Food Science



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Food Science Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Food Science				
Food Science Lessons Nature Walks & Scouting: Grades 9-12	Food Science Lessons	Food Science Lessons	Food Science Lessons Nature Notebook: Grades 9-12	Food Science Lessons Food Science Labs



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



Quick Links

(No Quick Links Assigned)

[Click THIS text
or scan the QR
code for links.](#)



Food Science Labs

How To Approach



Introduce

- Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.
- Relevant concept(s) are introduced in the text.
- Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.



Lab Procedure

- Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.
- Record all data and observations in the lab notebook.



Analysis & Conclusions

- After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.
- If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.

Term 1

WEEK 1

60m Food Science Labs - Lesson 1

Materials: Salt, Fat, Acid, Heat, ingredients for Caesar Salad p.48, protein, bread, indicated kitchen supplies

→ KITCHEN LAB

We'll learn more about salads in the coming weeks, but for this week, make Caesar Salad as on p.48-49 in SFAH. You may either use store-bought mayo for now or try the recipe (visual p.86-87 or traditional p.375). Serve alongside your choice of protein and bread/croutons. If you want to make your own mayonnaise, you might find the video below helpful.

∞ Video Link: Mayonnaise and the Science of Emulsions

WEEK 2

60m Food Science Labs - Lesson 2

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Salt Lesson. Decide on next Salt Lesson p.428 and add to shopping list, if you haven't already.

WEEK 3

60m Food Science Labs - Lesson 3

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Salt Lesson. Decide on next Salt Lesson p.428 and add to shopping list, if you haven't already.

WEEK 4

60m Food Science Labs - Lesson 4

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Salt Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 5

60m Food Science Labs - Lesson 5

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 6

60m Food Science Labs - Lesson 6

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 7

60m Food Science Labs - Lesson 7

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 8

60m Food Science Labs - Lesson 8

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 9

60m Food Science Labs - Lesson 9

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

Food Science Labs

[Click THIS text or scan the QR code for links.](#) QR

Term 1

WEEK 10 60m Food Science Labs - Lesson 10

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 11 60m Food Science Labs - Lesson 11

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 12 60m Food Science Labs - Lesson 12

Exam: Dinner Night

→ EXAMS

Answer exam questions from:
Kitchen Lab experience

Food Science Labs

[Click THIS text or scan the QR code for links.](#) 

Term 2

WEEK 1	 60m Food Science Labs - Lesson 13  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 2	 60m Food Science Labs - Lesson 14  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 3	 60m Food Science Labs - Lesson 15  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 4	 60m Food Science Labs - Lesson 16  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies Prep: Read Teacher Tip → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	★ TEACHER TIP Students are conducting citizen science next week and will not have their regular dinner night. They may need to adjust their schedule, so that their Kitchen Waste Audit is completed at the end of the week.
WEEK 5	 60m Food Science Labs - Lesson 17  Materials: internet access, kitchen scale → DATA COLLECTION Conduct the second Kitchen Waste Audit. Adjust your schedule, if needed, so that this audit is completed after all 4 Daily Food Waste Logs. ∞ Website Link: Food Conservation	
WEEK 6	 60m Food Science Labs - Lesson 18  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 7	 60m Food Science Labs - Lesson 19  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 8	 60m Food Science Labs - Lesson 20  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 9	 60m Food Science Labs - Lesson 21  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	

Food Science Labs

[Click THIS text or scan the QR code for links.](#) QR

Term 2

WEEK 10 60m Food Science Labs - Lesson 22

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 11 60m Food Science Labs - Lesson 23

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 12 60m Food Science Labs - Lesson 24

Exam: Dinner Night

→ EXAMS

Answer exam questions from:
Kitchen Lab experience

Term 3

WEEK 1  60m Food Science Labs - Lesson 25  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 2  60m Food Science Labs - Lesson 26  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 3  60m Food Science Labs - Lesson 27  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 4  60m Food Science Labs - Lesson 28  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 5  60m Food Science Labs - Lesson 29  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 6  60m Food Science Labs - Lesson 30  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 7  60m Food Science Labs - Lesson 31  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 8  60m Food Science Labs - Lesson 32  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	
WEEK 9  60m Food Science Labs - Lesson 33  Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies → KITCHEN LAB Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.	

Food Science Labs

[Click THIS text or scan the QR code for links.](#) QR

Term 3

WEEK 10 60m Food Science Labs - Lesson 34

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 11 60m Food Science Labs - Lesson 35

Materials: Salt, Fat, Acid, Heat, indicated ingredients, and kitchen supplies

→ KITCHEN LAB

Prepare the chosen Cooking Lesson. Decide on the next Cooking Lesson and add to the shopping list, if you haven't already.

WEEK 12 60m Food Science Labs - Lesson 36

Exam: Dinner Night

→ EXAMS

Answer exam questions from:
Kitchen Lab experience